
Using rDCM models in generative embedding to develop sex/gender-disaggregated tools for the diagnosis of ASD

Project report for
Translational Neuromodeling

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Abstract The present project builds upon a recent publication from Supekar et al. [1], which identified key brain regions for discriminating between female and male subjects diagnosed with autism spectrum disorder (ASD). Regression dynamic causal modeling (rDCM) is applied to “Resting-state” functional magnetic resonance imaging (rs-fMRI) data to extract the effective connectivity parameters between these brain regions. Classifiers are trained on these connectivity parameters separately for female and male subjects to build a sex/gender-disaggregated computational tool for ASD diagnosis. The trained classifiers are evaluated for their prediction accuracies and statistical significance on same and opposite sex/gender data. The classifier with the highest average accuracy rate, logistic regression trained on female connectivity matrices, predicts female ASD diagnosis with a balanced accuracy of 67% (p-value of 0.01) and male ASD diagnosis with a balanced accuracy of 54%. This difference in accuracies is significant with a p-value of 0.07 and demonstrates that using sex/gender-disaggregated data can improve the accuracy of computational diagnostic tools for ASD. The features used in this project are furthermore mechanistically interpretable and could lead to additional insight into which connectivity networks are significant for (sex/gender-disaggregated) ASD diagnosis. Last but not least, our work provides a fully automated and easily scalable pipeline for further studies; starting with selecting subjects and downloading their resting state fMRI data from the preprocessed ABIDE dataset, then extracting connectivity parameters using rDCM and finally using this data to train diagnostic classifiers.

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1 Motivation

Autism spectrum disorder (ASD) is a neurodevelopmental disorder which affected as many as one in 44 children aged 8 years old in the United States in 2018 and was about 4 times more prevalent among boys than among girls [2], with similar ratios reported by multiple other nation-wide studies [3–9]. It has been suggested that this sex/gender difference may be caused in part by ascertainment bias, underrecognition of females due to behavioral differences and issues of today’s diagnostic instruments[10], with criteria currently defined in ICD and DSM both being derived from descriptions of the behavior and development of affected adolescent boys[11, 12]. Building diagnostic tools which take into account and mitigate these biases is important two-fold:

- On average, females are not only diagnosed less frequently than males but also later. Earlier diagnosis and treatment of ASD is important and has been shown to improve later language and cognitive abilities and ameliorate the core symptoms.[13]
- Most autism studies include participants based on the reported male-female ratio. This implies our general understanding of ASD being substantially biased towards males.[10]

A recent study from Supekar et al.[1], which set out to determine whether neurobiology differs in females and males with ASD, has shown robust sex/gender differences in functional brain organization and their links to clinical symptoms in autism spectrum disorder using a novel spatiotemporal deep neural network (stDNN). These differences were namely different connectivity patterns in brain regions including motor, language and visuospatial attention systems, and could explain sex/gender particularities in clinical symptoms.

There already exist efforts to develop computational assays[14] for autism spectrum disorder which use fMRI data and the ABIDE data set. This project builds upon the results of the aforementioned study [1] and aims to construct a sex/gender-disaggregated computational assay as a tool for ASD diagnosis. For this, generative embedding is used: rDCM is applied to extract the effective connectivity parameters between the previously identified distinct brain regions, using data from the preprocessed ABIDE data set[15]. These features are then used to train classifiers on female and male subjects separately to predict whether they belong to the ASD or control group. Differences in prediction accuracies from these distinct models would be further evidence of sex/gender differences in the neurobiology of ASD patients, and could be the base for more efficient diagnostics tools.

2 Methods

All code used for this project was written in Python and MATLAB. The entire code as well as a list of requirements for running is available on the Git Repository (https://gitlab.ethz.ch/tn_projects_fs2022/project_group_5).

2.1 Selecting the data

The data utilized in the present study was taken from the ABIDE Preprocessed data set, which is composed of structural and resting state functional MRI images from both healthy individuals and individuals diagnosed with ASD[15]. The phenotypic information about each tested subject is also provided, as well as quality ratings of the obtained data. Of the four preprocessing pipelines which the ABIDE Preprocessed data set offers, that of the Data Processing Assistant for Resting-State fMRI (DPARSF) was chosen. This tool is based on SPM and most closely resembles the preprocessing done by Supekar et al. [1] by performing the following steps: After segmentation, realignment and co-registration, the images are normalized to the standard Montreal Neurological Institute (MNI152) template. The body motion is regressed out, but while Supekar et al. use 6 motion parameters, DPARSF uses the Friston 24-parameter model[16]. Both Supekar et al. and the pipeline used here apply a 6-mm Gaussian kernel for spatial smoothing and apply a band-pass filter ranging from 0.01 to 0.1 Hz. Finally, the ABIDE Preprocessed data set offers the option of global signal regression, which was decided against as this was not part of Supekar et al.’s preprocessing either.

After having determined the preprocessing, the subjects from the ABIDE data set that would be used for our models were chosen by filtering through the metadata. Among all the participants, subjects were

first discarded when their functional images' quality assessments made by manual ABIDE raters were not judged satisfactory, and when their mean frame wise displacement was greater than 0.5, once again to match the criteria mentioned by Supekar et al. [1] We further restricted the subjects to the ones not taking any medication, to exclude any effects this could have on the brain connectivity. More male subjects were left after the filtering, and to avoid having to tackle multiple class imbalances (male vs. female and ASD vs. control), they were matched to the female ones based on the site of data acquisition, their clinical diagnosis (ASD or control) and their age. This resulted in a total of 174 subjects (87 males, 87 females), 44 of which with ASD (22 males, 22 females). The subjects were thus selected, downloaded from ABIDE's S3 bucket and unzipped using a Python script.

2.2 Extracting the time-series of the ROIs

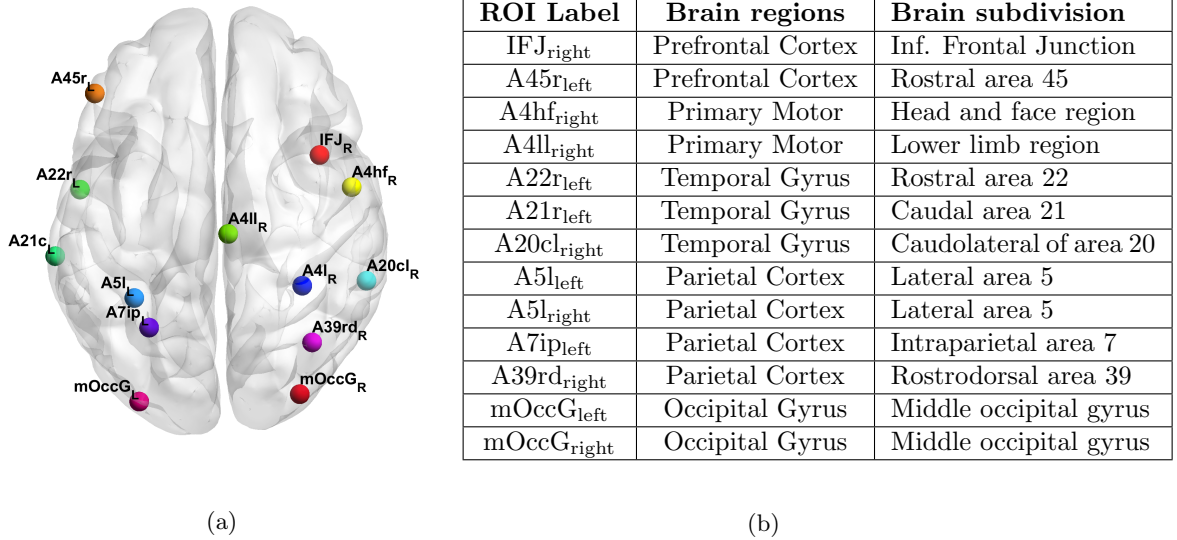


Figure 1: (a) Locations of the 13 Brainnetome ROIs used for the rDCM network and (b) the brain regions they come from. The brain networks were visualized with the BrainNet Viewer [17]

For each subject, the time-series of the regions of interest (ROIs) were extracted using SPM12 (version 7771, [18]) which automatically computes the first principal component of the time-series from all voxels within the ROI. The selected regions are ones defined by the Brainnetome atlas[19] and are the 13 areas mentioned by Supekar et al. [1] as "having contributed most significantly to predicting the ASD female class label". To benefit from the Volume of Interest time-series extraction function of SPM, a general linear model (GLM) had to be applied to each functional image. ABIDE Preprocessed provides resting-state for which the body motion has already been regressed out, with the used motion regressors not being specified. Therefore, the GLM used to extract the time series contains neither conditions nor regressors. A whole-brain mask was applied to the image along with the ROI masks when extracting the time series, to assure only data from within the actual brain was being acquired.

2.3 Creating the rDCM

The goal of this project is to determine whether there are significant and diagnostically exploitable differences in the effective connectivity, that is, the directed interactions among selected brain regions, of the networks of male and female patients with ASD. The method used to assess such connectivity is Dynamic Causal Modeling (DCM), a generative modeling framework of the neuronal states and how these give rise to the measured (fMRI) data [20]. However, the DCM implementation in SPM12 is restricted to small networks, allowing at most 8 ROIs in the default implementation to keep model inversion computationally feasible. To calculate the connectivity parameters between all 13 specified ROIs, the extension of DCM to regression DCM (rDCM) was used, as it allows efficient computation for very large networks. The analysis was performed using the corresponding toolbox [21] [22], with the open-source code available as part of the TAPAS software collection [23]. To use the resting-state functionality of rDCM validated in [24], no input was specified into the rDCM structure. Furthermore, no arguments were

passed either, so as to make a fully connected network by default. The rDCM connectivity parameter estimation were computed both for no sparsity constraints and no sparsity constraints. These endogenous connectivity matrices serve as the features on which in the next step the classifiers were trained.

2.4 Classification methods

To predict the diagnosis of the subjects based on their connectivity matrices, three different classifiers were used, namely logistic regression (with an l1 penalty for feature selection), a random forest classifier and an AdaBoost classifier. While logistic regression is a commonly used supervised learning algorithm, random forest[25] and AdaBoost[26] are more sophisticated ensemble methods which consist of sets of independently trained classifiers. This project is based on a relatively small data set, which poses a challenge to training an accurate estimator. The two ensemble methods were chosen to counteract these challenges, as they implement bagging and boosting methods which are used to reduce overfitting or bias, respectively [27]. Besides being small, the data set also suffers from a class imbalance, with control subjects outnumbering ASD subjects 3:1. To mitigate this issue, a balanced accuracy score, which weights samples inversely proportional to their relative class prevalence and random oversampling of the training data were used.

For each kind of classifier, two models were created and fit to the endogenous connectivity matrices of a specific sex/gender computed both with and without sparsity constraints along with the corresponding ASD labels. Each data set was beforehand split into training and testing subsets, to measure the accuracy of the classification on data never seen during training. It was noticed that the classification accuracies on the test set highly fluctuated depending on the specific train-test-split. To compensate for this variability, a repeated cross-validation scheme was implemented and the obtained balanced accuracy scores were then averaged over all folds and repeats. For the ensemble classifiers (Random Forest and AdaBoost), the scores also depended on the random initialization parameters of each model instance, which is why these models were even fit and averaged multiple times for each cross validation period. For all classifiers, stratification was used to ensure that the folds would preserve the same percentage of samples (before oversampling) for each class, to address the class imbalance between ASD and control subjects.

3 Results

3.1 Examining connectivity parameters

In a first step to evaluate the implications of considering sex/gender differences for ASD diagnosis, the endogenous connectivity matrices obtained from rDCM were analyzed. Four matrices were computed by averaging the ones of females and males with and without ASD (see the first two matrices of Fig. 2 and Fig. 3). The difference of the averaged matrices with and without ASD were then calculated for each sex/gender, with the goal of finding a significant absolute difference in connectivity strengths which could implicate single connections with a relevance for ASD diagnosis (see the third matrix of Fig. 2 and Fig. 3). To assess the significance of the differences, a permutation test was performed by averaging over the same sex/gender but with randomly permuted ASD labels to create a population of 1000 difference matrices. The p-value for each connections was calculated, using the formula $p = \frac{C+1}{n+1}$, where C corresponds to the number of permutations whose connection differences were stronger than the true ones and n is the total number of permutations. This calculation was furthermore corrected for the number of individual tests using the Bonferroni correction, and for two-sided testing, as the absolute difference was considered. Therefore, any significance level becomes $\alpha' = p/(169 \cdot 2) = \alpha/338$.

Furthermore, an additional matrix corresponding to the difference between sexes/genders of the difference between ASD label was computed (see Fig. 4). Its appearance informs us on the potential of this study. Should it be a null matrix, this would indicate no difference in the diagnostically relevant information contained in the endogenous connectivity matrices. On the other hand, should there be a structure in the matrix, using the connectivity matrices to train sex/gender-disaggregated ASD classifiers would have a potential for success. This pipeline was repeated for both with and without sparsity constraints when computing the rDCM, and the figures shown here are for the case without sparsity constraints.

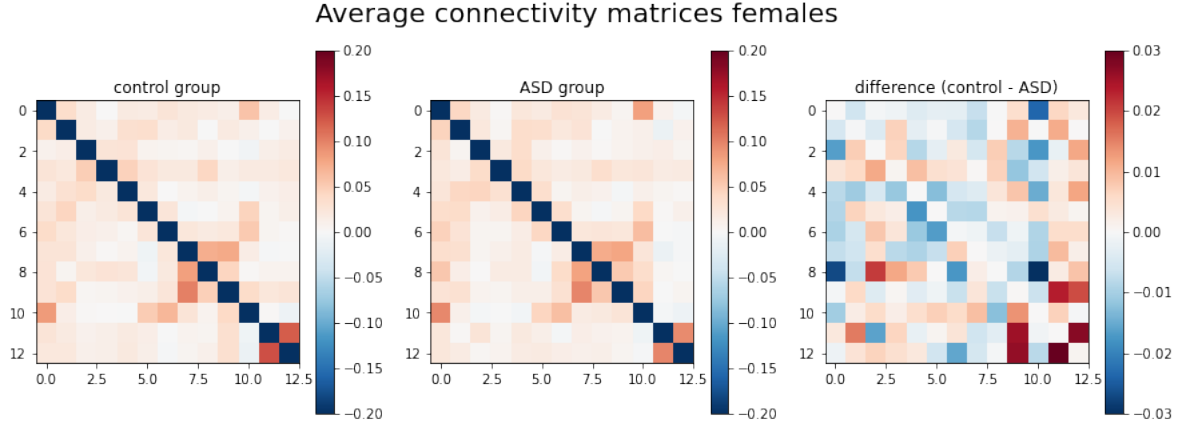


Figure 2: Endogenous connectivity matrices for rDCM without sparsity constraints, averaged over all females with ASD and all females without ASD. The third matrix shows the absolute value of the difference between the two first matrices.

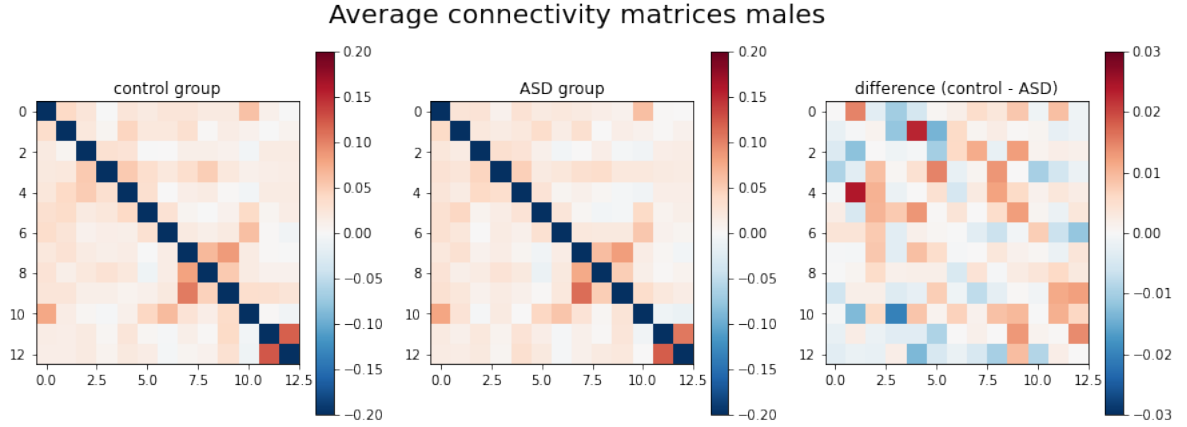


Figure 3: Endogenous connectivity matrices for rDCM without sparsity constraints, averaged over all males with ASD and all males without ASD. The third matrix shows the absolute value of the difference between the two first matrices.

The first observation that can be made from the resulting matrices is that the one displayed in Fig. 4 is not empty, but rather shows relatively strong differences in connections between sexes/genders. A similar result was obtained for the case with sparsity constraints. This thus validates the hypothesis that connectivity matrices between the selected ROIs have a potential of improving predictions for the ASD label when distinguishing females from males.

Given a corrected significance level of $\alpha' = 0.05/338 = 1.48 \cdot 10^{-4}$, none of individual connections in either sex/gender difference matrix (see right-most matrix in Fig. 2 3) are statistically significant. This being said, there is still some relevant information to be extracted here. While the lowest p-values for the difference matrices of both sexes/genders computed with sparsity constraints and for male subjects computed without sparsity are two orders of magnitude larger than α' , the lowest p-value for the female difference matrix computed without sparsity constraints is of the same order of magnitude as α' . These differences could potentially indicate the female connectivity matrices computed with sparsity restraints to contain more diagnostically relevant information. Moreover, one could argue that the 169 connections are not fully independent as they correspond to both directions of 78 connections and to self-connections which could even be ignored in significance tests from the start. Applying this softer correction to the significant threshold, would lead to one connection (the one connecting the lateral area 5 to the intraparietal area 7 in the left parietal cortex) in the female no-sparsity connectivity matrix to be significant for at least $\alpha = 0.10$.

That being said, while perhaps no clear single connection significantly differs between the groups, the overall average endogenous connectivity patterns carry at least some exploitable information.

Difference of avg females - males

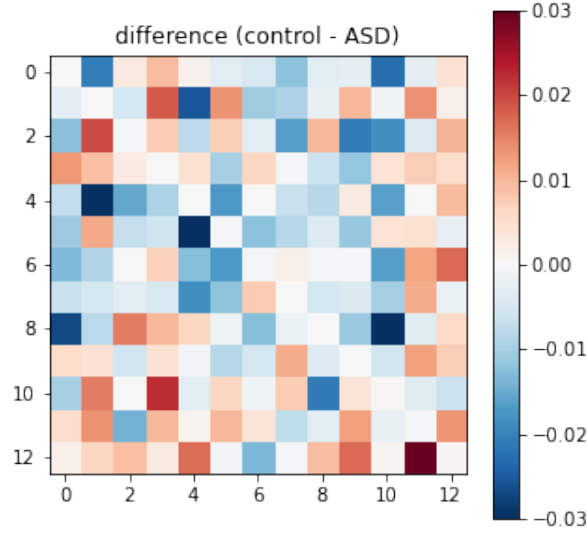


Figure 4: Difference of the third matrix of Fig. 2 and the third matrix of Fig. 3.

3.2 Classification results

Next, we sought out to leverage the connectivity data by using it to train different classifiers with the goal of predicting the ASD diagnosis for males and females. As the focus of this study is on the differences between the two sexes/genders, each type of classifier was trained separately on female and male data. To assess the quality of the prediction and the sex/gender specificity of the trained classifiers, four different accuracies were calculated for each new classifier instantiated at each fold of the cross-validation:

- Balanced prediction accuracy of a classifier trained and tested on female data
- Balanced prediction accuracy of a classifier trained on female data and tested on male data
- Balanced prediction accuracy of a classifier trained and tested on male data
- Balanced prediction accuracy of a classifier trained on male data and tested on female data

These scores were then averaged over the folds to give one accuracy per train-test split, which were then averaged over the splits to give one accuracy per prediction type. Histograms showing the distributions of the accuracies over the folds of each of the repeated train-test splits and aggregated over all train-test splits have been plotted, to show the variability that comes from the new instance of the classifier initialized at each fold, and from the random train-test split. The histograms of Fig. 5 show the distribution of accuracies for two of the five train-test split, whereas the one of Fig. 6 combines all five histograms.

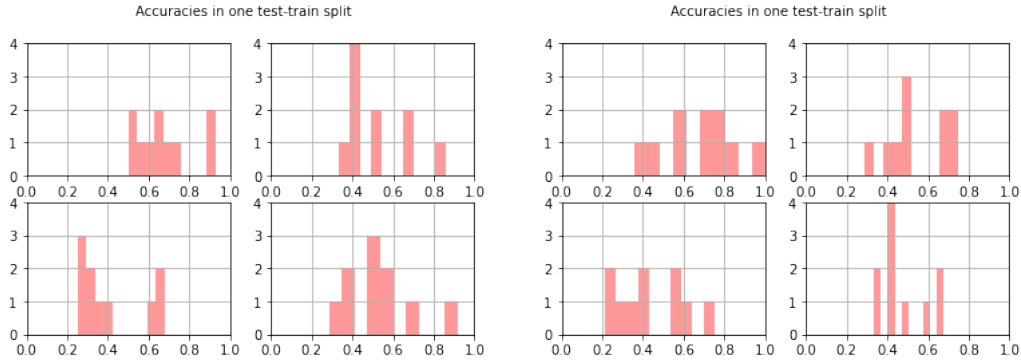


Figure 5: Histograms of the accuracy values obtained for two different cross-validation folds. Upper left: female trained - female tested; upper right: female trained - male tested; lower left: male trained - male tested; lower right: male trained - female tested

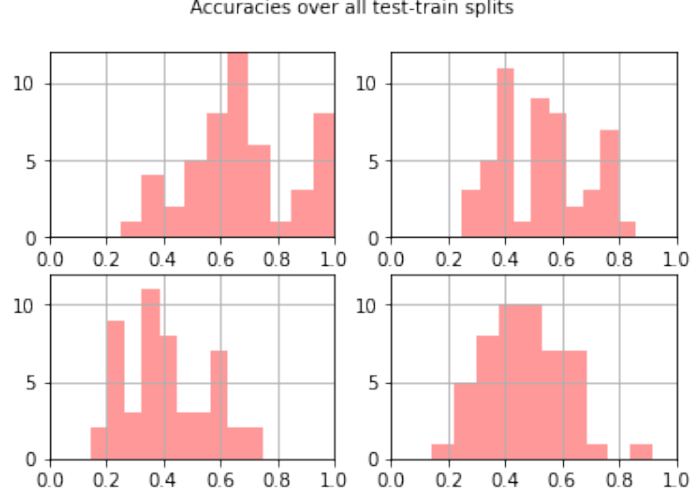


Figure 6: Histograms of the accuracy values obtained for one train-test split. Upper left: female trained - female tested; upper right: female trained - male tested; lower left: male trained - male tested; lower right: male trained - female tested

The desired output of the four different accuracy scores is to show that the prediction of a model trained and tested on the same sex/gender is more accurate in predicting the ASD label than a model trained on one sex/gender and tested on the other.

To certify the significance of the accuracies and ensure that the results were not obtained by chance, permutation testing was implemented by randomly shuffling the ASD labels of both male and female subjects one hundred times. The above accuracy scores were computed for each permutation, and compared to the non-permuted case. The ensuing p-values of each prediction were calculated, using the above formula without Bonferroni correction, and where C now corresponds the number of permutations whose accuracy score was higher than the true score.

Furthermore, the significance of the difference of prediction accuracies of a specific model on test data of the same sex/gender as its training data versus test data of the opposite sex/gender was also computed. The p-values were obtained according to the above formula, where C is now the number of permutations where the difference between these accuracies is higher than that of the non-permuted case. Tab. 2 summarizes all the interesting p-value results for the three classifiers considered in this study and for both without and with sparsity constraints during rDCM. Tab. 1 shows the accuracies for the five lowest p-values above.

To ease the visualization of the results, histograms showing the distribution of the accuracies obtained with a specific classifier and constraint specification during permutation testings have been plotted. A vertical line indicates the accuracy for the corresponding non-permuted test (see an example in Fig. 7, obtained with Logistic Regression and no sparsity constraints).

Train/Test	Log. Regression		Random Forest		Ada Boost	
	No sparse	Sparse	No sparse	Sparse	No sparse	Sparse
Fem / Fem	0.67	-	0.53	-	0.60	-
Fem/(Fem-Mal)	0.13	-	-	-	0.12	-

Table 1: Accuracies and accuracy differences, respectively, corresponding to the lowest p-values below

Train/Test	Log. Regression		Random Forest		Ada Boost	
	No sparse	Sparse	No sparse	Sparse	No sparse	Sparse
Fem / Fem	0.00990	0.90099	0.06931	0.30693	0.05941	0.48515
Mal / Mal	0.70297	0.54455	0.84158	0.50495	0.47525	0.95050
Fem/(Fem-Mal)	0.06931	0.85149	0.14851	0.53465	0.09901	0.56436

Table 2: p-values of the prediction accuracy of classifier trained and tested on the same sex/gender, and of the difference in accuracy between trained-tested on female and trained on female, tested on male

Permuted accuracies LogReg nosparse

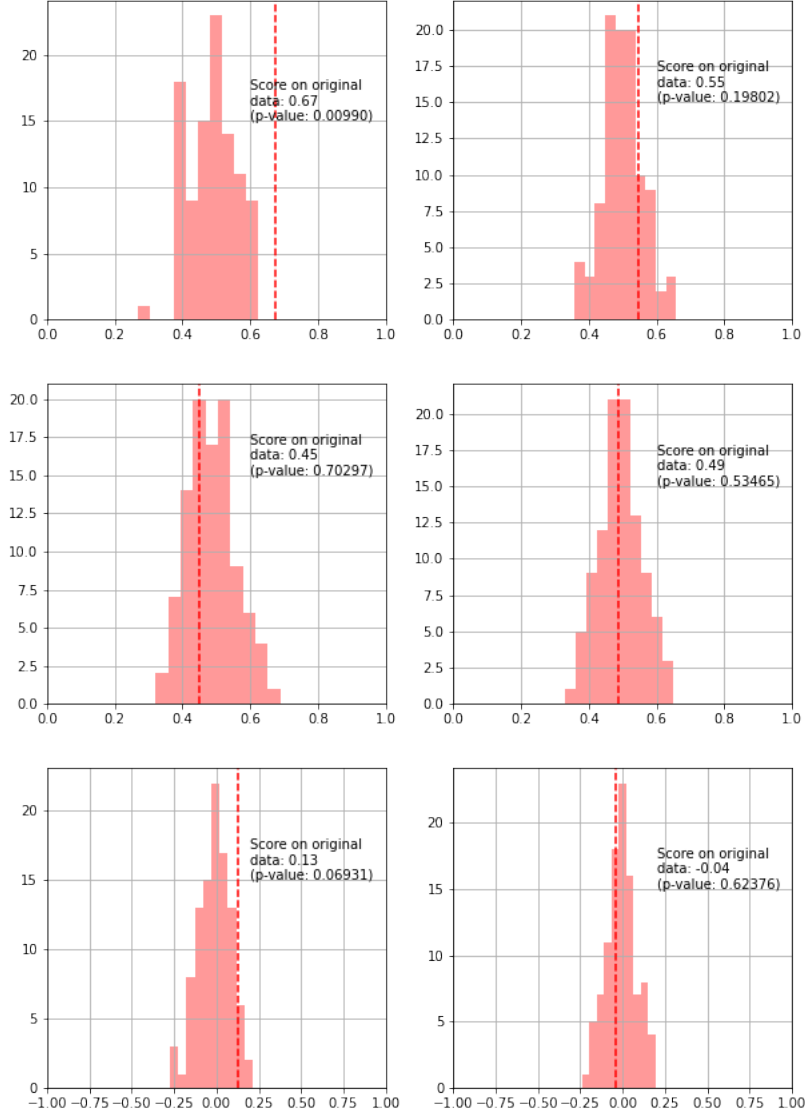


Figure 7: Histogram of the distribution of the accuracies obtained during permutation testings with Logistic Regression and no sparsity constraints. Upper left: female trained - female tested; upper right: female trained - male tested; middle left: male trained - male tested; middle right: male trained - female tested; lower left: female trained - difference female/male tested; lower right: male trained - difference male/female tested

To start with, the predictions of the ASD label for female subjects were correct with a maximal accuracy of 67% for the Logistic Regression classifier without sparsity constraints (see Fig. 7 and Tab. 1). This is a competitive value compared to other recent studies on the ABIDE dataset ([14], [28], [29]). Furthermore, the permutation testing in that case shows that such a value for the female ASD predictions is highly significant, as it resulted in a p-value of less than 1% (see Tab. 2). Given the formula for the p-value calculation, this in fact means that over all the permutations, none could perform better than the non-permuted case. This result is generalisable to all the classifiers in the no-sparsity constraints setting, as they all have a prediction accuracy greater than 50% with significant or close to significant p-values. It is interesting to notice that the Random Forest Classifier performs more poorly than the other two in classifying females. Furthermore, such an accuracy is specific to predicting females, as the results obtained when testing on males with any classifier trained with females have a poor accuracy

(13% difference compared to when tested with female data, as shown in Tab. 1) and are consistently insignificant. The p-value contained in the last line of Tab. 2 shows that the calculation of this difference is significant.

On the other hand, none of the classifiers allows to make correct predictions for male subjects (even when trained on males), as the accuracies are always in the range of 45-50%. It was also noticed that overall, the prediction accuracy values of males-trained classifiers are not different when tested on male data or female data. Finally the p-values show no significance in any of the results obtained with males-trained classifier, which leads to the conclusion that the classifiers are not able to extract any meaningful data from the male subjects' connectivity matrices.

It is also interesting to mention that classifiers trained on female data obtained with sparsity constraints during rDCM perform consistently more poorly than those without the constraints. This result is in accordance with the ones obtained for the connection strengths of the endogenous connectivity matrices.

4 Discussion

The most compelling finding this project presents is that of a significant difference between applying a classifier trained on female data to predict the diagnosis of female versus that of male subjects. The classifier with the highest average accuracy rate, Logistic Regression trained on female connectivity matrices computed with no sparsity constraints, predicts female ASD diagnosis with a balanced accuracy of 67% and male ASD diagnosis with a balanced accuracy of 54%. This difference in accuracies is significant with a p-value of 0.07 and demonstrates that using sex/gender-disaggregated data can improve the accuracy of computational diagnostic tools for ASD. Furthermore, the average accuracy of our logistic regression classifier is within the range of accuracy scores reported by recently published studies that also used the ABIDE data set, which lay between 65%-75% [14][28][29]. While these other works all use convolutional neural networks (CNNs) to extract features and make predictions from the rsfMRI data, this project uses generative embedding, and first extracts effective connectivity matrices using rDCM based on brain regions indicated to mark sex/gender differences in ASD patients. Thus, the features used in this project are mechanistically interpretable and could lead to additional insights into which connectivity networks are significant for (sex/gender-disaggregated) ASD diagnosis.

On the downside, only our classifiers which were trained on female data deliver any significant or accurate predictions. Those trained on male data performed very poorly across the board, with no significant difference between predicting female or male ASD and no significant predictions in general. This may be attributed to the way the ROIs upon which the rDCM was built were chosen, as these were specifically those "having contributed most significantly to predicting the ASD female class label"[1]. Choosing another set of brain regions could potentially improve the male ASD classifier. This would open up a whole new topic which is worth investigating, namely developing separate connectivity models for the sexes/genders and evaluating them using Bayesian model selection. Another way to improve the male classifiers would be to include more male subjects in the training. These were limited in this project to match the number of available female subjects. Thus, including additional data sets with more female subjects would lead to a larger amount of usable data in general.

Being able to use larger data sets for both female and male classifiers would also help stabilize prediction accuracies which currently fluctuate strongly and require computationally expensive averaging processes. In terms of classification methods which could potentially improve predictions in general, there is also much left to explore with noteworthy additional possibilities including support vector machines and CNNs.

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